

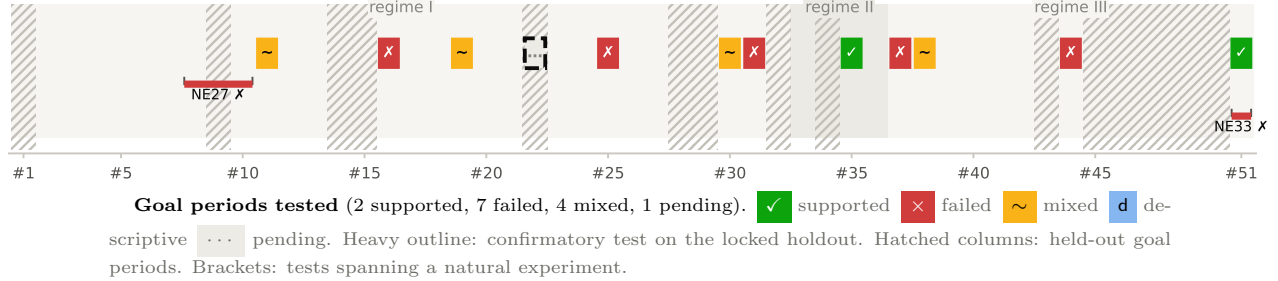
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 0.5/5 Usefulness 2/5

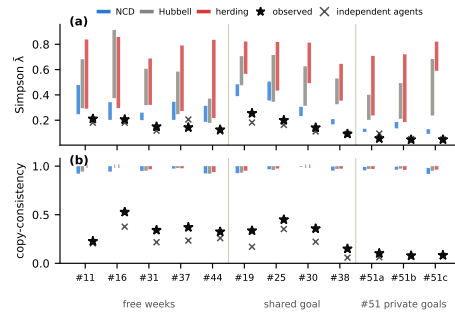
Bottom line. Free LLM agents state private sub-tasks but work on shared repos when one exists; no cooperative (NCD) dynamics, neutral copying at best, and copying follows what agents read.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Agent slots carry a project label per 30-min window: a cluster of the agent's self-written goal, the repo it links, or (round 1b) the repo it commits to. Exact finite- N simulations with the real label gaps fit neutral cooperative dynamics (NCD), Hubbell drift and a herding rival to 9 statistics. **NCD failed** in free weeks (4/5 failed, 1 mixed), and in rounds 1 and 1b all three models are rejected on stated goals in every free week (joint check $p < 0.01$): 72–91% of goal clusters are held by one agent. Round 1b (second embedding model, style removed) leaves this intact, but a control week whose project is known (#35, one fork per room) shows stated goals just as fragmented while work sits 96–100% on the room's fork: goal clusters measure sub-tasks, not projects. Work labels concentrate on shared repos where one exists; there Hubbell-like copying can fit (#30), without NCD's signature. Copying follows reading: a project named in a message the agent has read predicts a switch (OR 8.6 vs unread).



Observables. Per period, goal clusters (km24). (a) Observed Simpson λ (star) against the 95% predictive ranges of the three fitted copying models (bars), and an independent-agents null (cross; each agent's day sequence shifted). The observation sits below all three models and close to the null. (b) Share of project switches that join a project another agent currently holds. Every copying model predicts ≈ 1 ; free and shared weeks give 0.15–0.53, and #51 (private roles) 0.08–0.10.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Neutral cooperative dynamics doesn't apply to these swarms. What agents *say* they are doing is a spread of private sub-tasks in every week, so stated-goal abundances are a poor project census. What they *work* on concentrates on a shared repo when the task offers one, and spreads out when the task hands each agent its own artifact or role (#44 free room, #51), as H11 found for effort. There, neutral copying is a fair description; there is no cooperator core. The copying that exists runs through reading, so an operator who wants a shared project must assign or announce it. Diagnostics: work-label singleton fraction and copy-consistency.

PHYSICAL MODEL

Model 06 (Piñero et al. 2025). N individuals (agent slots), each holding one species (project). Per step, with probability μ a new project replaces a random slot C . Otherwise C copies slot A 's project only if A and a third slot B hold different projects: cooperation, so rare projects recruit more per capita. $\lambda = \sum_i x_i^2$ is the Simpson index of shares $x_i = n_i/N$. Rivals: Hubbell (always copy) and herding (copy favoured when A, B agree).

$$b_n = (1 - \mu) \frac{n}{N} \left(1 - \frac{n}{N}\right)^2, \quad P_n \propto \frac{(1 - \mu)^n}{n} e^{-(n - N\lambda^*)^2/2N}$$

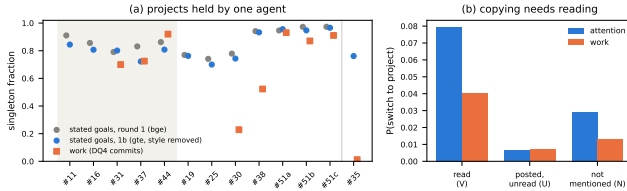
$$\frac{1}{\mu} = \sqrt{\frac{\pi N}{2}} \operatorname{erfcx}\left(\sqrt{\frac{N}{2}}(\mu - \lambda^*)\right), \quad \mu_B = \frac{e^{-2}}{\sqrt{2\pi N}}, \quad \mu_L = \sqrt{\frac{2}{\pi N}}$$

$$P(\text{recruit to } i) \propto x_i e^{\beta x_i} : \quad \beta < 0 \text{ NCD}, \quad 0 \text{ Hubbell}, \quad > 0 \text{ herding}$$

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 NCD beats rivals (free)	4 failed, 1 mixed → same count (#11, #16 swap)	failed
Adequacy, stated goals	all 3 models rejected in 5/5 free weeks (both rounds)	failed
Singletons, goals (free)	0.79–0.91 → 0.72–0.84 (gte, style removed)	robust
P2 rare-project advantage	#31 $\hat{\beta}$ $-0.78 \rightarrow +1.47$; sign varies by embedding	failed
Work vs goals (R1b-3a)	work more concentrated: $\bar{\lambda}$ 5/5, supported copy 4/5	supported
NCD on work (R1b-3b)	inadequate 4/5; #30 NCD and Hubbell adequate, no core	supported
Copying needs reading	OR read vs unread 8.6 (attention), 5.6 (work)	supported
#35 known universe (native)	work 96–100% on own fork; goal singletons 0.76	supported
#51 rivals share (native)	goals 2/3 blocks; attention 0/3; 1/3 work 0/3	
#44 assigned vs free (native)	assigned arm more concentrated in 3 spaces	mostly



Second figure. (a) Share of projects held by one agent: stated goals (grey round 1, blue round 1b) stay high everywhere, including #35 where the project is known; work labels (orange) drop where a shared repo exists (#30, #35, #38). Shaded: free weeks. (b) Probability of switching to a project held by another agent, by whether a message naming it had been read, posted but not yet read, or neither.

GOAL PERIODS IN DETAIL

G22	...	1b: pending (confirmatory, not run) (holdout)
G11	~	1b: mixed (P1 by rule; all models still inadequate)
G16	×	1b: failed (km24 only; all models inadequate)
G19	~	1b: mixed (P6 unchanged; all models inadequate)
G25	×	1b: failed (P6: conformist no longer beats NCD)
G30	~	1b: mixed (P6); work labels: NCD and Hubbell adequate...
G31	×	1b: failed (P1 6/6; work labels also inadequate; P2 n...
G35	✓	1b: supported (native: known universe recovered; goal...
G37	×	1b: failed (km24 only; all models inadequate)
G38	~	1b: mixed (P6; km24 all inadequate; work: Hubbell ade...

+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Compute was scarce: the full estimator (all clusterings) re-ran on #11 and #31 only; elsewhere the primary clustering, work labels and model-free statistics. Likelihood ratios near ± 2 and $\hat{\beta}$ depend on the embedding. In two-room weeks the “unread” class is mostly the other room’s messages. The day-shift null cannot tell a persistent shared field (room assignment) from copying. Work labels cover public commits only (dense from #30).

ROUND 2

NCD/Hubbell with agent-specific fields on work labels; repo-anchored stated goals; #22 confirmation (now on work labels as a secondary) after sign-off.

No round-2 redirects yet.

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